C13.1 About the interface control signals

This section lists the signals that are available to configure interface behavior in components that support flexible interfaces.

Note

The signals in this section are optional, and are not part of the AMBA ACE protocol. However, if used, it is an architectural requirement that all interface control signals are stable, and remain static, on reset.

The AMBA ACE configuration signals are:

BROADCASTINNER

When asserted, indicates that the interface must broadcast Inner Shareable transactions. If this signal is asserted, **BROADCASTOUTER** must also be asserted.

BROADCASTOUTER

When asserted, indicates that the interface must broadcast Outer Shareable transactions.

BROADCASTCACHEMAINT

When asserted, indicates that the interface must broadcast cache maintenance operations to downstream caches.

This signal controls the broadcast of cache maintenance operations and is asserted whenever a downstream cache exists below the coherent interconnect. Asserting this signal results in all CleanShared, CleanInvalid, and MakeInvalid transactions being broadcast.

When this signal is deasserted, whether cache maintenance operations are broadcast depends on:

- their shareability domain
- **BROADCASTINNER** and **BROADCASTOUTER** settings.

Table C13-1 shows the valid combinations of the interface control signals and the corresponding transactions that are broadcast.

Table C13-1 Interface control signals

Signal			Transactions broadcast
BROADCAST INNER	BROADCAST OUTER	BROADCAST CACHEMAINT	
0	0	0	None
0	1	0	All Outer Shareable
1	1	0	All Inner Shareable and Outer Shareable
0	0	1	All cache maintenance operations
0	1	1	• All Outer Shareable • All cache maintenance operations
1	1	1	• All Inner Shareable and Outer Shareable • All cache maintenance operations

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